

# RAMESWAR BHATTACHARJEE

Electronic Structure Theory •  $\pi$ -Conjugated and Radical Materials • Machine Learning for Chemical Discovery

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## RESEARCH PROFILE

Computational chemist developing predictive electronic-structure and machine-learning frameworks for electronically complex molecular and materials systems. My research connects orbital symmetry, topology, and spin/charge delocalization to structure–property relationships in  $\pi$ -conjugated polymers, open-shell radical assemblies, and catalytic reaction networks. Recent independent work establishes design principles for topological conjugated polymers, quantifies non-classical “pancake” bonding in radical  $\pi$ -stacks, and introduces interpretable descriptor-based machine learning that predicts DFT-level electronic properties across thousands of radical dimer packings.

*I am seeking a tenure-track faculty position in which to build an independent program on predictive design of open-shell and topological organic electronic materials, combining high-level electronic structure theory with data-driven discovery.*

## RESEARCH METRICS

|                            |           |         |                         |                      |                        |
|----------------------------|-----------|---------|-------------------------|----------------------|------------------------|
| 37                         | 1,180     | 19      | 17                      | 7                    | 3                      |
| Peer-reviewed publications | Citations | h-index | First / co-first author | Corresponding author | Journal cover features |

Citation metrics from Google Scholar, August 2026. Six papers as both first and corresponding author; one additional paper as designated co-first author (equal contribution).

## ACADEMIC APPOINTMENTS

|                                                                                                                                                                                      |                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Research Assistant Professor</b>                                                                                                                                                  | June 2025 – Present   |
| Georgetown University, Department of Chemistry — independent research on topological conjugated polymers, radical $\pi$ -stacking, and machine-learning-guided materials prediction. |                       |
| <b>Postdoctoral Fellow</b>                                                                                                                                                           | Nov. 2022 – May 2025  |
| Georgetown University — with Prof. Miklos Kertesz. Electronic structure of $\pi$ -conjugated polymers and open-shell radical aggregates.                                             |                       |
| <b>Postdoctoral Fellow</b>                                                                                                                                                           | Dec. 2020 – Oct. 2022 |
| University of South Dakota — with Prof. Pere Miró. Actinide speciation and polyoxocationic cluster chemistry.                                                                        |                       |
| <b>Postdoctoral Fellow</b>                                                                                                                                                           | Jan. 2019 – Dec. 2020 |
| University of Delaware — with Prof. Dion Vlachos and Dr. Stavros Caratzoulas. Computational catalysis for biomass valorization (DOE Energy Frontier Research Center).                |                       |

## EDUCATION

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Ph.D., Chemistry</b>                                                                                                                                                    | 2018 |
| Indian Association for the Cultivation of Science (IACS), Kolkata. Advisor: Prof. Ayan Datta.<br>Thesis: Computational Studies of Homogeneous and Heterogeneous Catalysis. |      |
| <b>M.Sc., Chemistry</b>                                                                                                                                                    | 2012 |
| Indian Institute of Technology (IIT) Guwahati.                                                                                                                             |      |
| <b>B.Sc., Chemistry</b>                                                                                                                                                    | 2010 |
| University of Burdwan.                                                                                                                                                     |      |

## FEDERAL RESEARCH FUNDING — CONTRIBUTING PERSONNEL

Named project participant on active federal awards, as reported to the funding agencies by the Principal Investigators.

|                                                                                                                     |                        |
|---------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>NSF CHE-2505194</b>                                                                                              | Sept. 2025 – Aug. 2028 |
| Collaborative Research: Radical and Biradical Polycyclic Aromatic Hydrocarbons in Complex Aggregation Environments. |                        |

Division of Chemistry, Macromolecular/Supramolecular/Nanochemistry Program. PI: M. Kertesz (Georgetown); collaborative with H. Lischka (Texas Tech). Total award: \$323,702.

- Role: NSF-reported project participant. **First author on two of the four publications formally reported as scientific outputs of this award** (*ACS Mater. Au* 2025; *ACS Phys. Chem. Au* 2026).

#### DOE DE-SC0019017

2023 – Present

Exploration of Radial Conjugation Pathways in  $\pi$ -Electron Materials.

U.S. Department of Energy, Office of Science. PI: J. D. Tovar (Johns Hopkins University).

- Role: DOE-reported project participant. Corresponding author on resulting publication, *Chem. Sci.* 2025, 16, 14595.

#### Independent research proposals authored

- Ramanujan Fellowship, Government of India (submitted 2026) — independent faculty research program on computationally guided design of conjugated materials for optoelectronic applications.
- Alexander von Humboldt Fellowship (2022) — theoretical strategies for selective heavy-metal separation from groundwater systems.

## SELECTED RESEARCH CONTRIBUTIONS

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### Topological electronic structure in $\pi$ -conjugated materials

- *J. Am. Chem. Soc.* 2024 — Strain-induced topological transition in  $\pi$ -conjugated polymers
- *Chem. Sci.* 2025 — Quinonoid radial  $\pi$ -conjugation
- *Chem. Mater.* 2024 — Continuous bandgap tuning through mechanical strain

Established electronic topology, quinonoid–aromatic competition, and mechanical strain as tunable design parameters for controlling band structure and emergent electronic phases in polymeric semiconductors.

### Radical $\pi$ -stacking and non-classical pancake bonding

- *J. Am. Chem. Soc.* 2024 — Charge effects in  $\pi$ -stacked perylenes
- *Chem. Sci.* 2024 — Trimeric triphenylene radical cation aggregation and bonding
- *ACS Mater. Au* 2025 — Pancake bonding in cationic acene dimers

Quantified how charge distribution, orbital symmetry, and spin/charge delocalization govern intermolecular stabilization in stacked radical assemblies, in sustained collaboration with crystallographic groups that validated the predictions experimentally.

### Machine learning for structure–property prediction in open-shell materials

- **Descriptor-based ML across 2,582  $\pi$ -radical dimer geometries** spanning five PAH radical systems, linking molecular packing to DFT-derived electronic structure.
- Nineteen chemically interpretable geometric descriptors predict fragment-orbital SOMO coupling, frontier-orbital gap, and interaction energies with test-set  $R^2 = 0.94$ – $0.98$  ( $0.82$ – $0.92$  under held-out geometry clusters).
- Validated against system-mean baselines and 500-fold Y-randomization controls, separating genuine geometric information from family-level correlation — a rigor standard often missing in chemical ML.
- Datasets and analysis code released openly (Zenodo DOI: 10.5281/zenodo.21712041).

Demonstrates that interpretable, physically grounded descriptors can replace expensive broken-symmetry DFT for rapid screening of radical packing motifs — the methodological foundation of my proposed independent program.

### Computational catalysis and reaction mechanism discovery

- *Green Chem.* 2020 — Biomass-derived lubricant base oil synthesis
- *Chem. Sci.* 2019; *Angew. Chem. Int. Ed.* 2016 — Metal-free CO<sub>2</sub> reduction chemistry
- *Chem. Eur. J.* 2017 — Direct and autocatalytic reductive elimination

Advanced mechanistic understanding of homogeneous and heterogeneous reaction landscapes relevant to sustainable synthesis and CO<sub>2</sub> transformation.

## COMPUTATIONAL EXPERTISE

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- **Molecular electronic structure:** Gaussian, ORCA, Q-Chem. Ground- and excited-state DFT, TDDFT, broken-symmetry DFT for open-shell and antiferromagnetically coupled systems, fragment-orbital DFT for electronic coupling and transfer integrals.
- **Periodic and solid-state:** VASP, CP2K, Quantum ESPRESSO. Band structure, density of states, strain-dependent electronic structure of one-dimensional polymers, surface adsorption.

- **Multireference methods:** COLUMBUS, MOLCAS/OpenMolcas, Molpro. Multireference treatments of biradical and polyradical character in open-shell PAHs, developed through collaboration with the Lischka group.
- **Conformational sampling and dynamics:** GOAT conformer search, GFN2-xTB semiempirical tight-binding, automated high-throughput generation and curation of multi-thousand-structure datasets.
- **Machine learning and data science:** Python, scikit-learn, PyTorch. Ensemble tree methods (extremely randomized trees), multilayer perceptron regression, descriptor engineering, cross-validation and cluster-based validation design, permutation and impurity feature-importance analysis, Y-randomization controls.
- **Analysis and bonding theory:** Natural bond orbital analysis, energy decomposition, Mulliken/Löwdin population analysis, spin-density and pancake-bonding characterization.
- **High-performance computing:** Georgetown University High Performance Computing cluster. Routine management of large-scale parallel quantum-chemical workflows, including automated generation and analysis of multi-thousand-structure datasets.
- **Programming and workflow:** Python, Bash, Git; version-controlled and openly archived analysis pipelines.

## PEER-REVIEWED PUBLICATIONS

37 publications, reverse chronological order. Applicant name in bold. Asterisk (\*) indicates corresponding author; double dagger (‡) indicates equal contribution.

1. Reid, H.W.; **Bhattacharjee, R.\***; Sun, P.B.; Skala, M. E.; Serck, N. P.; Golder, M. R.\*; Kertesz, M.; Jasti, R.\*. High Molecular Weight [n]Cycloparaphenylene Conjugated Polymers and Copolymers with Improved Electron Delocalization. *Macromolecules* 2026, 59, 6963–6972. DOI: 10.1021/acs.macromol.6c00539
2. **Bhattacharjee, R.\***; McCormack, M.;‡ Zhu, Y.; Wei, Z.; Kertesz, M.\*; Petrukhina, M.A.\* Chemical Oxidation of Coronene: A Crystallographic and Computational Study of Charge-Regulated  $\pi$ -Stacking. *ACS Phys. Chem. Au* 2026, 571-579. DOI: 10.1021/acspyschemau.6c00020
3. Yin, G.\* **Bhattacharjee, R.**; Wang, T.; Kertesz, M. Wilson polygons and the topology of zero-dimensional systems. *Phys. Rev. B* 2026, 113, 115104. DOI: 10.1103/7wx3-6syf
4. McCormack, M.; **Bhattacharjee, R.**; Wei, Z.; Petrukhina, M.A.\* Chemical Oxidation of Chrysene: A Structural and Theoretical Description of a Mixed-Valent Trimeric Radical Cation. *Cryst. Growth Des.* 2025, 25, 9549–9554. DOI: 10.1021/acs.cgd.5c01176
5. McCormack, M.;‡ **Bhattacharjee, R.**;‡ Wei, Z.; Kertesz, M.\*; Petrukhina, M.A.\* Building a Mixed Polyarene  $\pi$ -Stack through Chemical Oxidation: A Structural and Theoretical Investigation. *Precis. Chem.* 2025, 3, 631–639. DOI: 10.1021/prechem.5c00066
6. **Bhattacharjee, R.\***; Tovar, J.D; Kertesz, M.\* Quinonoid radial  $\pi$ -conjugation. *Chem. Sci.* 2025, 16, 14595. DOI: 10.1039/D5SC00639B
7. **Bhattacharjee, R.\***; Lischka, H.; Kertesz, M.\* Pancake Bonding in the Stabilization of Cationic Acene Dimers *ACS Mater. Au* 2025, 5, 365–376. DOI: 10.1021/acsmaterialsau.4c00153
8. **Bhattacharjee, R.\***; Kertesz, M.\* Topological transition in aromatic and quinonoid  $\pi$ -conjugated polymers induced by static strain. *J. Am. Chem. Soc.* 2024, 146, 26497–26504. DOI: 10.1021/jacs.4c10064
9. **Bhattacharjee, R.**;‡ McCormack, M.;‡ Jervis, H.; Wei, Z.; Petrukhina, M.A.\*; Kertesz, M.\* A unique trimeric triphenylene radical cation: Stacking aggregation, bonding, and stability. *Chem. Sci.* 2024, 15, 15221-15231. DOI: 10.1039/d4sc03774j
10. **Bhattacharjee, R.\***; Jervis, H.; McCormack, M.; Petrukhina, M.A.; Kertesz, M.\* Structure and Bonding in  $\pi$ -Stacked Perylenes: The Impact of Charge on Pancake Bonding. *J. Am. Chem. Soc.* 2024, 146, 10465–10477. DOI: 10.1021/jacs.3c14065
11. **Bhattacharjee, R.\***; Kertesz, M.\* Continuous Topological Transition and Bandgap Tuning in Ethynyl-Linked Acene  $\pi$ -Conjugated Polymers through Mechanical Strain. *Chem. Mater.* 2024, 36, 1395–1404. DOI: 10.1021/acs.chemmater.3c02547
12. Ren, L.; Xiao, Y.; **Bhattacharjee, R.**; Wu, J.; Tang, P.; Caratzoulas, S.; Meng, C.; Guo, Q.\*; Tsapatsis, M.\* High Yield of L-Sorbose via D-Glucose Isomerization in Ethanol over a Bifunctional Titanium-Boron-Beta Zeolite. *Chem. Eur. J.* 2024, 30, e202402341 DOI: 10.1002/chem.202402341
13. McCormack, M.; **Bhattacharjee, R.**; Jervis, H.; Wei, Z.; Kertesz, M.\*; Petrukhina, M.A.\* Structural and Computational Insights into Partially Oxidized  $\pi$ -stacked Perylene. *Cryst. Growth Des.* 2023, 23 7496-7503. DOI: 10.1021/acs.cgd.3c00912
14. **Bhattacharjee, R.**; Hare, E; Miró, P.\* Structure, Properties, and Reactivity of Polyoxocationic Zirconium and Hafnium Clusters: A Computational Investigation *Inorg. Chem.* 2023, 62, 5081-5087. DOI: 10.1021/acs.inorgchem.2c04142
15. **Bhattacharjee, R.**; Miró, P.\* Aqueous Speciation of Tetravalent Actinides in the Presence of Chloride and Nitrate Ligands *Inorg. Chem.* 2022, 37, 14718-14725. DOI: 10.1021/acs.inorgchem.2c02064

16. Deegan, M. M.; **Bhattacharjee, R.**; Caratzoulas, S.; Bloch, E. D.\* Stabilizing Porosity in Organic Cages Through Coordination Chemistry. *Inorg. Chem.* 2021, 60, 7044– 7050. DOI: 10.1021/acs.inorgchem.0c03590
17. Liu, S; **Bhattacharjee, R.**; Li S; Danielson, A; Mazal, T; Saha, B.\*; Vlachos, D.G.\* Thiol-promoted catalytic synthesis of high-performance furan-containing lubricant base oils from biomass-derived 2-alkylfurans and ketones. *Green Chem.* 2020, 22, 7896-7906. DOI: 10.1039/D0GC01897J
18. Rowland, C. A; Lorzing, G. R; **Bhattacharjee, R.**; Caratzoulas, S; Yap, G; Bloch, E. D.\* Design and synthesis of aryl-functionalized carbazole-based porous coordination cages. *Chem. Commun.* 2020, 56, 9352-9355. DOI: 10.1039/D0CC03910A
19. Anju, B. S.; **Bhattacharjee, R.**; Ahammad, S.; Datta, A.\*; Kundu, S.\* Deoxygenation of Nitrosoarene by N- Heterocyclic Carbene (NHC): Elusive Breslow-type Intermediate Bridging Carbene and Nitrene. *Chem. Commun.* 2020, 56, 12166-12169. DOI: 10.1039/D0CC05192F
20. Lorzing, R. G.; Gosselin, E. J.; Lindner, B. S.; **Bhattacharjee, R.**; Yap, G.; Caratzoulas, S.; Bloch, E. D.\* Design and synthesis of capped-paddlewheel-based porous coordination cages. *Chem. Commun.* 2019, 55, 9527-9530. DOI: 10.1039/C9CC05002G
21. **Bhattacharjee, R.**; Majumdar, T.; Datta, A.\* Analysis of pseudo jahn–teller distortion based on natural bond orbital theory: Case study for silicene. *J. Comput. Chem.* 2019, 40, 1488-1495. DOI: 10.1002/jcc.25815
22. Sau, S. C.; **Bhattacharjee, R.**; Vardhanapu, P. K.; Hota, P.K.; Vijaykumar, G.; Govindarajan, R.; Datta, A.\*; Mandal, S. K.\* Transforming atmospheric CO<sub>2</sub> into alternative fuels: a metal-free approach under ambient conditions. *Chem. Sci.* 2019, 10, 1879–1884. DOI: 10.1039/C8SC03581D
23. Rath, S. P.; Sengupta, D.; Ghosh, P.; **Bhattacharjee, R.**; Chakraborty, M.; Samanta, S.; Datta, A.; Goswami, S.\* Effects of Ancillary Ligands on Redox and Chemical Properties of Ruthenium Coordinated Azoaromatic Pincer. *Inorg. Chem.* 2018, 57, 11995–12009. DOI: 10.1021/acs.inorgchem.8b01558
24. **Bhattacharjee, R.**; Datta, A.\* Understanding Thermal and Photochemical Aryl-Aryl Cross-coupling using metal complex. *Chem. Eur. J.* 2018, 24, 13636-13646. DOI: 10.1002/chem.201802634
25. Pramanick, R.; **Bhattacharjee, R.**; Sengupta, D.; Datta, A.\*; Goswami, S.\* An azoaromatic ligand as four electron four proton reservoir: catalytic dehydrogenation of alcohols by its Zinc(II) complex. *Inorg. Chem.* 2018, 57, 6816-6824. DOI: 10.1021/acs.inorgchem.8b00034
26. **Bhattacharjee, R.**; Datta, A.\* Role of Carbon support for subnanometer gold-cluster catalyzed disiloxane synthesis from hydrosilane and water. *J. Phys. Chem. C* 2017, 121, 20101-20112. DOI: 10.1021/acs.jpcc.7b04419
27. **Bhattacharjee, R.**; Nijamudheen, A.; Datta, A.\* Direct and autocatalytic reductive elimination from gold complexes ([ (Ph<sub>3</sub>P)Au(Ar)(CF<sub>3</sub>)(X)], X= F, Cl, Br, I): The key role of halide ligands. *Chem. Eur. J.* 2017, 23, 4169-4179. DOI: 10.1002/chem.201605784
28. **Bhattacharjee, R.**; Datta, A.\* Supported sub-nanometer gold cluster catalyzed transfer hydrogenation of aldehydes to alcohols. *J. Phys. Chem. C* 2016, 120, 24449-24456. DOI: 10.1021/acs.jpcc.6b07520
29. **Bhattacharjee, R.**; Nijamudheen, A.; Karmakar, S.; Datta, A.\* Strain control: Reversible H<sub>2</sub> activation and H<sub>2</sub>/D<sub>2</sub> exchange in Pt complexes. *Inorg. Chem.* 2016, 55, 3023-3029. DOI: 10.1021/acs.inorgchem.5b02904
30. Sau, S. C.; **Bhattacharjee, R.**; Vardhanapu, P. K.; Vijaykumar, G.; Datta, A.\*; Mandal, S. K.\* Metal-free reduction of CO<sub>2</sub> to methoxyborane under ambient conditions through borondiformate formation. *Angew. Chem. Int. Ed.* 2016, 128, 15371-15375. DOI: 10.1002/anie.201609040
31. Asha, K.; **Bhattacharjee, R.**; Mandal, S.\* Complete transmetalation in a metal–organic framework by metal ion metathesis in a single crystal for selective sensing of phosphate ions in aqueous media. *Angew. Chem. Int. Ed.* 2016, 128, 11700-11704. DOI: 10.1002/anie.201606185
32. Sengupta, D.; **Bhattacharjee, R.**; Pramanick, R.; Rath, S. P.; Saha Chowdhury, N.; Datta, A.\*; Goswami, S.\* Exclusively ligand-mediated catalytic dehydrogenation of alcohols. *Inorg. Chem.* 2016, 55, 9602-9610. DOI: 10.1021/acs.inorgchem.6b01310
33. Roy, S.; Baral, A.; **Bhattacharjee, R.**; Jana, B.; Datta, A.; Ghosh, S.; Banerjee, A.\* Preparation of multi-coloured different sized fluorescent gold clusters from blue to NIR, structural analysis of the blue emitting Au<sub>7</sub> cluster, and cell- imaging by the NIR gold cluster. *Nanoscale* 2015, 7, 1912-1920. DOI: 10.1039/C4NR04338C
34. Pratik, S. M.; Chowdhury, C.; **Bhattacharjee, R.**; Jahiruddin, S.; Datta, A.\* Pseudo Jahn–Teller distortion for a tricyclic carbon sulfide (C<sub>6</sub>S<sub>8</sub>) and its suppression in S-oxygenated dithiine (C<sub>4</sub>H<sub>4</sub>(SO<sub>2</sub>)<sub>2</sub>). *Chem. Phys.* 2015, 460, 101-105. DOI: 10.1016/j.chemphys.2015.04.009
35. **Bhattacharjee, R.**; Nijamudheen, A.; Datta, A. Mechanistic insights into the synergistic catalysis by Au (I), Ga (III), and counterions in the Nakamura reaction. *Org. Biomol. Chem.* 2015, 13, 7412-7420. DOI: 10.1039/C5OB00626K
36. Nijamudheen, A.; **Bhattacharjee, R.**; Choudhury, S.; Datta, A.\* Electronic and chemical properties of germanene: The crucial role of buckling. *J. Phys. Chem. C* 2015, 119, 3802–3809. DOI: 10.1021/jp511488m

37. Maiti, D. K.; **Bhattacharjee, R.**; Datta, A.\*; Banerjee, A.\* Modulation of fluorescence resonance energy transfer efficiency for white light emission from a series of stilbene-perylene based donor-acceptor pair. *J. Phys. Chem. C* 2013, 117, 23178-23189. DOI: 10.1021/jp409042p

## MANUSCRIPTS UNDER REVIEW AND IN PREPARATION

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1. **Bhattacharjee, R.\***; Lischka, H.; Kertesz, M.\* How Molecular Packing Controls Electronic Structure in  $\pi$ -Stacked Radical Dimers: A DFT and Descriptor-Based Machine-Learning Study. *Submitted to Journal of Chemical Information and Modeling*, 2026. Preprint posted. Data and analysis code openly archived: Zenodo DOI 10.5281/zenodo.21712041.
2. **Bhattacharjee, R.\***; Lischka, H.; Kertesz, M.\* Mapping the  $\pi$ - $\sigma$  Bonding Landscape in a Pancake-Bonded Radical Dimer via Data-Driven and Chemistry-Guided Analysis. *Manuscript in final preparation*, 2026.

## CONFERENCE PRESENTATIONS

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- Machine learning of packing-electronic structure relationships in  $\pi$ -stacked radical dimers. *Gordon Research Conference on Computational Materials Science and Engineering*, Sunday River, Newry, ME, Aug. 2-7, 2026. Poster.
- Topological conjugated polymers under mechanical strain and a prospect on machine learning. *Third International Symposium on Machine Learning in Quantum Chemistry*, Knoxville, TN, Oct. 5-8, 2025. Poster.
- Topological phases of conjugated polymers under mechanical strain. *ACS National Meeting, Fall 2025*, Washington, DC, Aug. 20, 2025. Oral.
- Mechanical and chemical tuning of the band gap of linearly bridged one-dimensional acene polymers. *ACS National Meeting, Fall 2023*, San Francisco, CA, Aug. 17, 2023. Oral.
- Speciation of  $\text{AnCl}_6^{2-}$  (An = U, Np, Pu) in aqueous solution. *ACS National Meeting, Spring 2022*, San Diego, CA, Mar. 20, 2022. Oral.

## TEACHING AND MENTORING

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### Teaching experience

#### Faculty Assistant — Physical Chemistry I

Georgetown University.

Fall 2025

#### Teaching Assistant — Computational Chemistry Methods

Georgetown University.

Spring 2025

#### Teaching Assistant — Physical Chemistry I

Georgetown University.

Fall 2024

### Courses prepared to teach

- **Undergraduate:** Physical Chemistry I and II (thermodynamics, kinetics, quantum chemistry), General Chemistry.
- **Graduate:** Computational Chemistry Methods, Electronic Structure Theory, Chemical Bonding and Group Theory.
- **Proposed new course:** Machine Learning for Molecular and Materials Discovery — a project-based course pairing electronic structure fundamentals with descriptor design, model validation, and reproducible open-science practice.

### Research mentoring

- Four undergraduate researchers mentored since 2019, across two institutions.
- Georgetown University (2022-present): mentored an undergraduate researcher to **two co-authored publications** in *J. Am. Chem. Soc.* and *Cryst. Growth Des.*; currently mentoring a second undergraduate researcher.
- University of Delaware (2019-2020): mentored two undergraduate researchers through the NSF Research Experiences for Undergraduates (REU) program.
- Training focus: electronic structure theory, quantum-chemical workflow design, high-performance computing practice, and reproducible computational analysis.

## AWARDS AND HONORS

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- EFRC Postdoctoral Fellowship, Catalysis Center for Energy Innovation (DOE Energy Frontier Research Center), University of Delaware, 2019-2020.
- Senior Research Fellowship, Council of Scientific and Industrial Research (CSIR), India, 2015-2018.
- Junior Research Fellowship, Council of Scientific and Industrial Research (CSIR), India, 2013-2015.
- CSIR National Eligibility Test (NET), Chemical Sciences — All-India Rank 44, 2012.
- Graduate Aptitude Test in Engineering (GATE), Chemical Sciences — All-India Rank 167, 2012.

- Three journal cover features.

## PROFESSIONAL SERVICE

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- Peer reviewer, **50+ manuscript reviews completed**, primarily for American Chemical Society and MDPI journals; additional reviews for Springer Nature and Wiley titles.
- Reviewer Board Member, *Inorganics* (MDPI).
- Postdoctoral Symposium Judge, Georgetown University.
- Member, American Chemical Society.
- Open science: public release of computational datasets and machine-learning analysis code via Zenodo.

## REFERENCES

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**Prof. Miklos Kertesz** — *Postdoctoral advisor and current supervisor*

Department of Chemistry, Georgetown University. kertesz@georgetown.edu

**Prof. Hans Lischka** — *Collaborator, NSF CHE-2505194*

Research Professor of Chemistry, Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, TX 79409. hans.lischka@ttu.edu

**Prof. Dionisios G. Vlachos** — *Postdoctoral advisor*

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**Prof. Ayan Datta** — *Doctoral advisor*

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